Pairwise Sequence Alignments

BioInf 2025/2026 1st Semester

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Tentative Schedule

- **1st test: 23/10/2025 (Thursday) 11:00**
- **2nd test: 12/12/2025 (Friday)**
- 1st project:
  - Announced: 9/10/2025
  - Submission: 23/10/2025
  - Discussions: 29/10/2025
- 2nd project:
  - Announced: 20/11/2025
  - Submission: 4/12/2025
  - Discussions: 10/12/2025

Summary

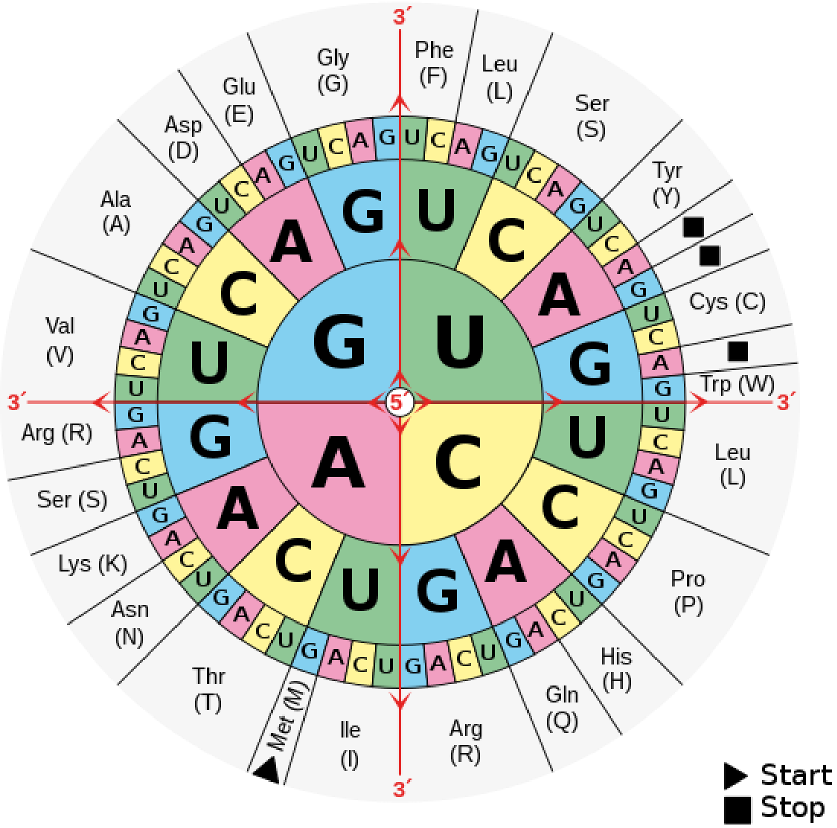
- Sequence alignment applications
- How to align sequences manually
- Manhattan tourist problem
- Needleman-Wunch Algorithm

Recap

- Biological data is sequential (DNA, RNA, proteins)
- Biological processes transform DNA->RNA->proteins
- Computers can process these sequences and simulate the biological processes
- Computers can find patterns in biological sequences

Genetic Code and Central Dogma

AAUGCUCGUAAUUAG
AAU-GCU-CGU-AAU-UUA → Asn-Ala-Arg-Asn-Leu
AUG-CUC-GUA-AUU-UAG → Met-Leu-Val-Ile-Stop
UGC-UCG-UAA-UUU → Cys-Ser-Stop-Stop



- Is this always true?
- In biology, it rarely is

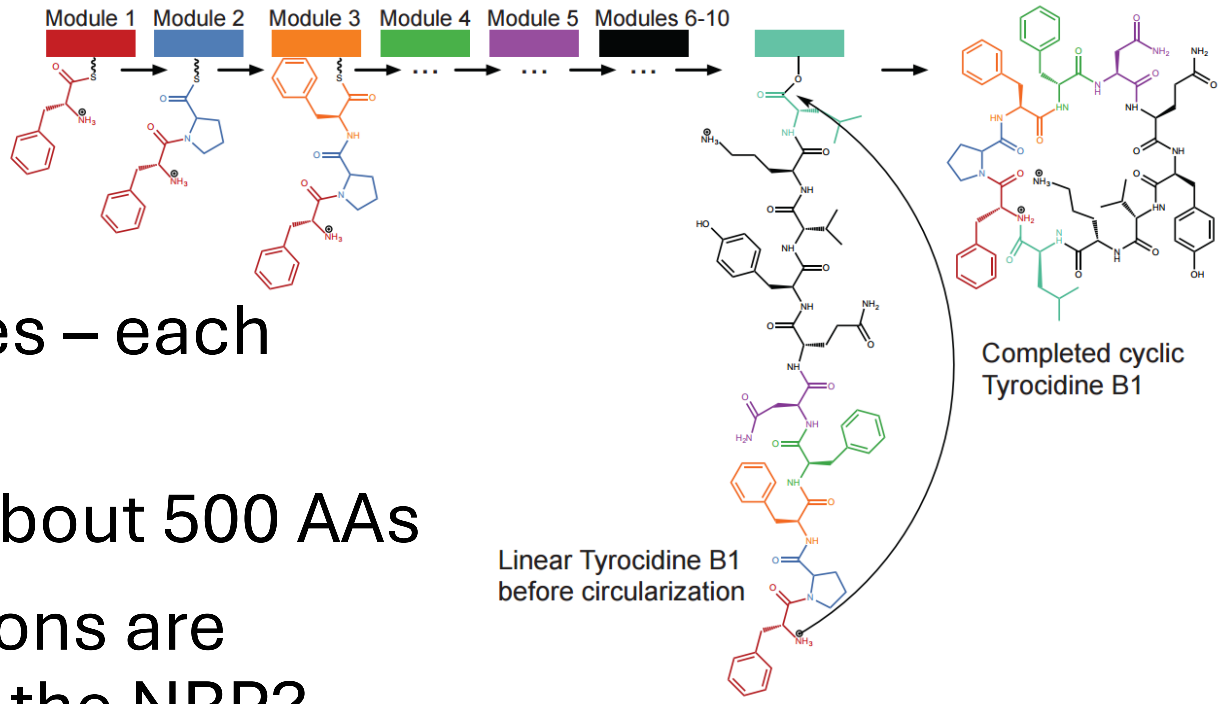
Non-ribosomal peptides

- Tyrocidine B1 is an antibiotic produced by *Bacillus brevis*
  - Peptide consisting of 10 Aas

Three-letter notation	Val-Lys-Leu-Phe-Pro-Trp-Phe-Asn-Gln-Tyr
One-letter notation	V K L F P W F N Q Y

  - We can't find the DNA sequence corresponding to this AA sequence in the *Bacillus brevis* genome!
- Non-Ribosomal Peptides (NRPs) are synthesized by NRP synthetase protein and not by ribosomes – does not require RNA
- DNA -> RNA -> Protein (NRP synthetase) -> NRP

NRPs



- NRP synthetase has 10 modules – each adds a single aminoacid
- Each module (A-domain) has about 500 AAs
- How can we know which positions are responsible for a specific AA in the NRP?
- Only three matching positions??

YAFDLGYTCMFPV**L**LG**G**GELHIVQKETYTAPDEIAHYI**K**EHGITYIKLTPSLFHTIVNTASFAFDANFESLRLIVLGGEKIIPIDVIAFRKMYGHTEFINHYGPTEATIGA
AFDVSAGDFARAL**L**TG**G**QLIVCPNEVKMDPASLYAIIK**K**YDITIFEATPALVIPLMEYIYEQKLDISQLQILIVGSDSCSMEDFKTLVSRFGSTIRIVNSYGVTEACIDS
IAFDASSWEIYAP**L**LN**G**GTVVCIDYYTTIDIKALEAVF**K**QHHIRGAMLPPALLKQCLVSAPTMISSLEILFAAGDRLSSQDAILARRAVGSGVYNAYGPTENTVLS

- What can we do?

NRPs

- What if we shift the second sequence by one position?

```
YAFDLGYTCMFPVLLGGGELHIVQKETYTAPDEIAHYIKEHGITYIKLTPSLFHTIVNTASFAFDANFESLRLIVLGGEKIIPIDVIAFRKMYGHTEFINHYGPTEATIGA
-AFDVSAGDFARALLTGGQLIVCPNEVKMDPASLYAIIKKYDITIFEATPALVIPLMEYIYEQKLDISQLQILIVGSDSCSMEDFKTLVSRFGSTIRIVNSYGVTEACIDS
IAFDASSWEIYAPLLNGGTVVCIDYYTTIDIKALEAVFKQHHIRGAMLPPALLKQCLVSAPTMISSLEILFAAGDRLSSQDAILARRAVGSGVYNAYGPTENTVLS
```

- Now we have 11 matches. A few more adjustments

```
YAFDLGYTCMFPVLLGGGELHIVQKETYTAPDEIAHYIKEHGITYIKLTPSLFHTIVNTASFAFDANFESLRLIVLGGEKIIPIDVIAFRKMYGHTE-FINHYGPTEATIGA
-AFDVSAGDFARALLTGGQLIVCPNEVKMDPASLYAIIKKYDITIFEATPALVIPLMEYI-YEQKLDISQLQILIVGSDSCSMEDFKTLVSRFGSTIRIVNSYGVTEACIDS
IAFDASSWEIYAPLLNGGTVVCIDYYTTIDIKALEAVFKQHHIRGAMLPPALLKQCLVSA----PTMISSLEILFAAGDRLSSQDAILARRAVGSGV-Y-NAYGPTENTVLS
```

- We can now say which parts are **conserved** (red) and which are specific to each domain

Cystic Fibrosis

- Genetic disorder – autosomal recessive (i.e. you can be a carrier without symptoms)
- European ancestry: 1/25 chance of being a carrier, affects 1/3,000 newborns
  - One of the most common rare diseases
- No known cure, but responsible gene has been found (CFTR)
  - However not everyone has the same mutation
- We need sequence alignment to identify candidate disease-causing mutations

Mutation	Frequency worldwide[180]
ΔF508	66–70%[20]
G542X	2.4%
G551D	1.6%
N1303K	1.3%
W1282X	1.2%
All others	27.5%

Sequence Alignment

- Mutations: subsequences can disappear (deletions) or be added (insertions) to the genome through evolution (a.k.a. indels)
- Comparing similar sequences of different genomes, the positions may not align perfectly – we need to find the best possible alignment
- This is a bad alignment:

```
ATGCATGC
TGCATGCA
```

This is a good alignment:

```
ATGCATGC–
–TGCATGCA
```

- How to decide?

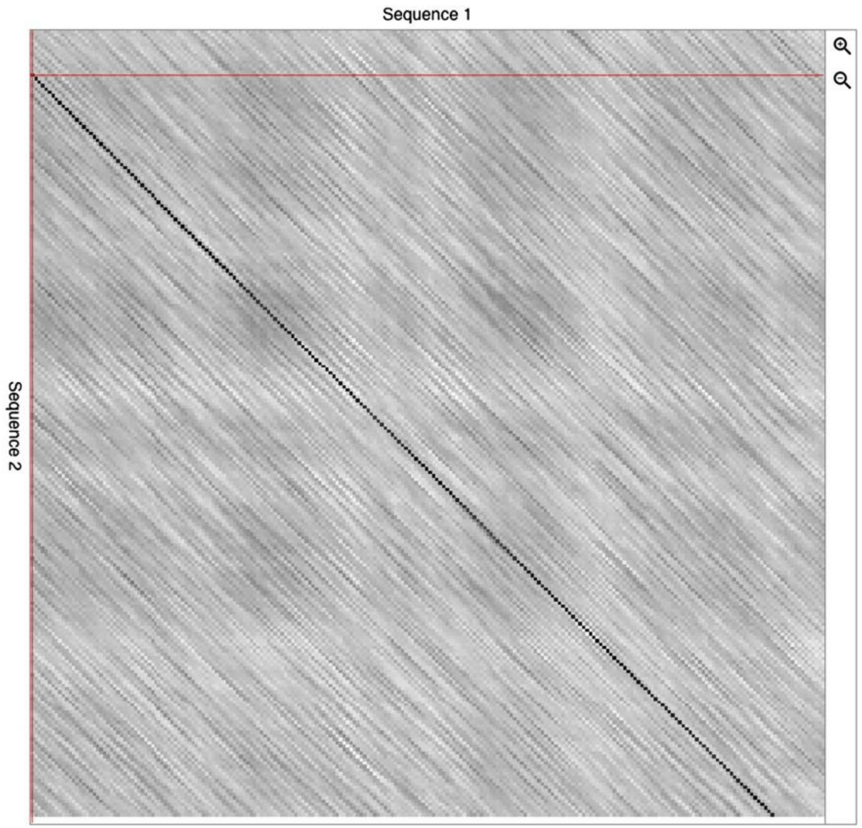
Sequence Alignment

```
A T - G T T A T A
A T C G T - C - C
```

- At each position:
  - 1 - align the symbols of each sequence at that position
  - 2 – add a gap to one of the sequences and align it with the symbol at the other sequence
- We want to maximize the number of matched position (where the symbol is the same) and minimize the number of mismatches (different symbols) and gaps (sometimes called indels)
  - Convention: gap at the top/first seq – insertions; bottom/second seq – deletion
- Longest common subsequence problem
- In summary: Where to put these gaps?

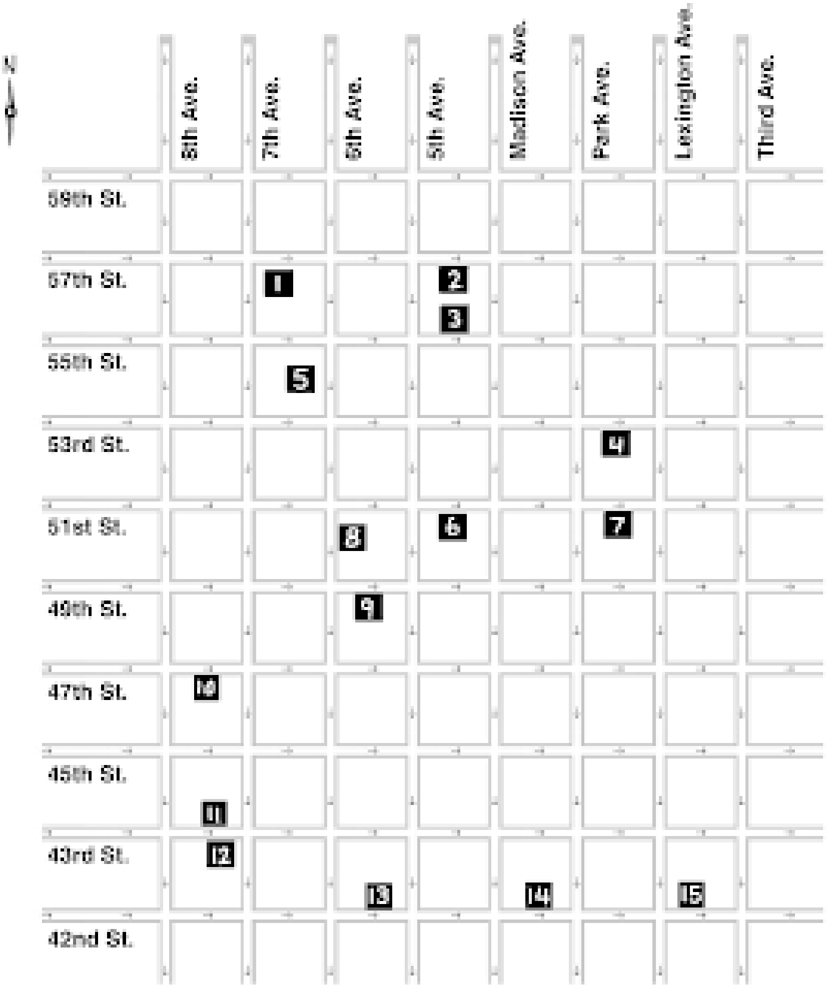
Visual Alignments – Dot Plots

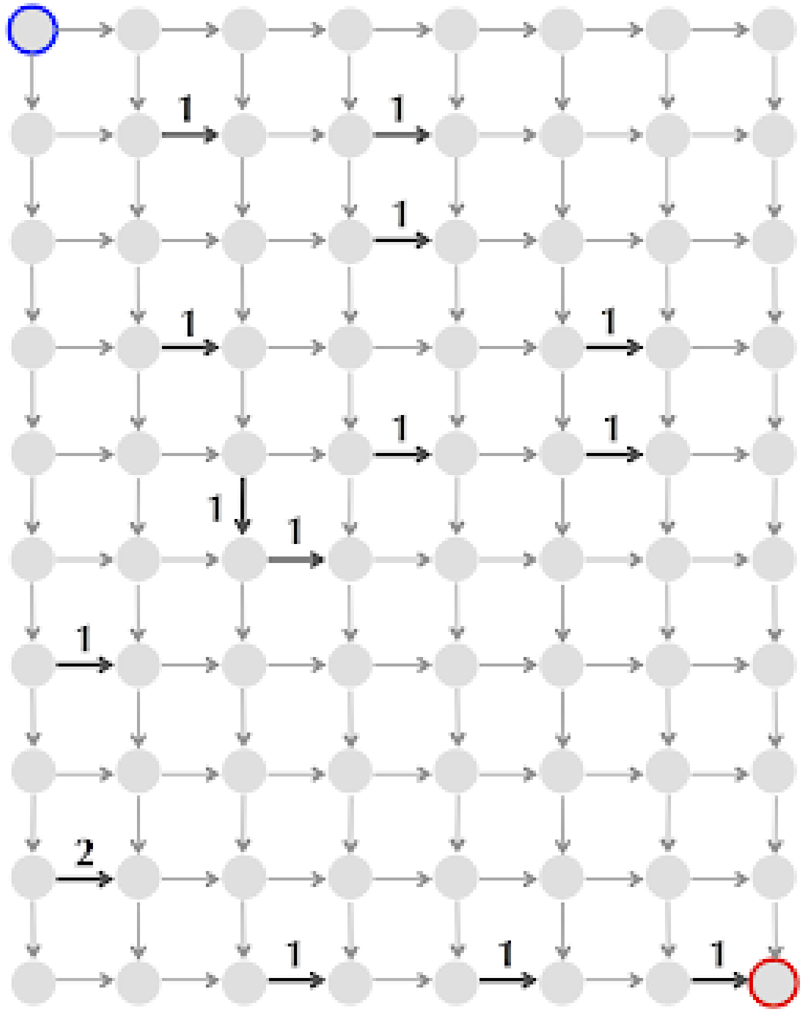
- Graphical representations of two sequences:
  - one sequence over the rows and the other over the columns.
- Highlight within the matrix regions of the sequences where the similarity is high.



- `dotlet` (https://dotlet.vital-it.ch) is a web-based application, to create dot plots from two protein sequences.

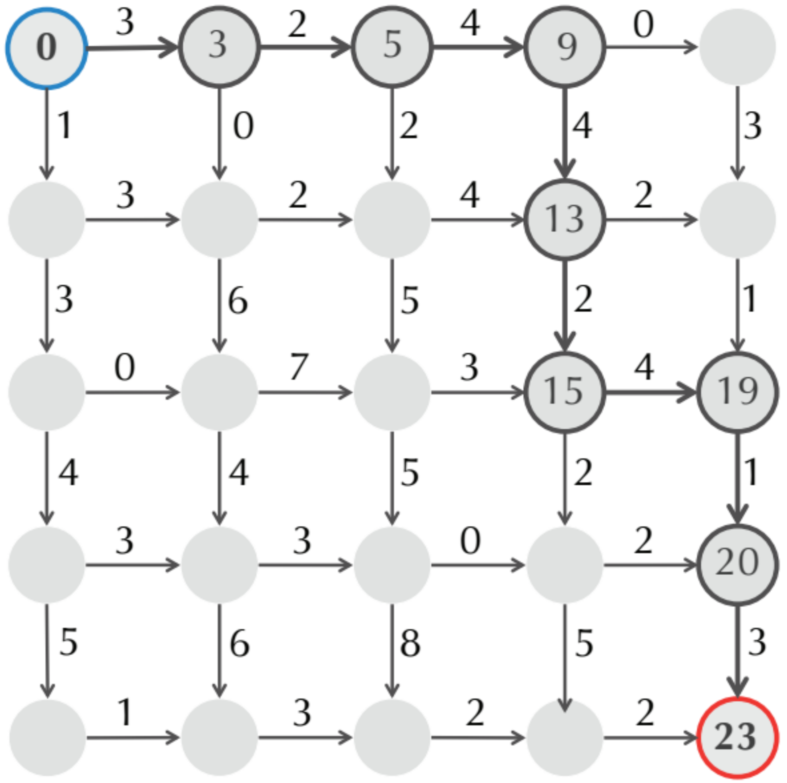
Manhattan Tourist Problem





Manhattan Tourist Problem

- Problem: find maximum weight path
  - (e.g. maximize number of attractions we visit)
- How many possible paths on a nxn grid?
- Brute force too slow
- Greedy approach might miss the global optima
- We can generalize this as finding the longest path in a directed acyclic graph (DAG)



Back to Sequence Alignment

- We can model Pairwise Sequence Alignment as a DAG
  - Each node is a partial solution
  - Each edge corresponds to a choice: match, add gap to first seq, add gap to second seq, which leads to different partial solution
  - Each partial solution has a score

0		1		2		2		3		4		5		6		7		8
	A		T		-		G		T		T		A		T		A	
	A		T		C		G		T		-		C		-		C	
0		1		2		3		4		5		5		6		6		7
	↘		↘		→		↘		↘		↓		↘		↓		↘	

Substitution Matrices and Gap Penalties

- How to score?
- Example (simplified):
  - Each gap, -2
  - Each identity, +1
  - Each difference, -1

```
  C T A T A G C A T T A G
  - - A T - G C A C T A G
 -2-2+1+1-2+1+1+1-1+1+1+1           Total: +1
```

Global Alignment

- Needleman-Wunsch Algorithm
  - The best alignment to A_{i-1}, B_{j-1} plus the match A_i, B_j
  - Or the best alignment to A_i, B_{j-1} and align B_j with gap
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- $S_{i,j} = max \begin{cases} S_{i-1,j-1} + sm(A_i, B_j) \\ S_{i,j-1} + g \\ S_{i-1,j} + g \end{cases}$

- Example: AGC | AAAC
  - $g = -2$
  - $sm(A_i, B_j) = \begin{cases} 1 & if \quad A_i = B_j \\ -1 & if \quad A_i \neq B_j \end{cases}$

	-	A	A	A	C
-	0	-2	-4	-6	-8
A	-2				
G	-4				
C	-6				

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-	0	-2	-4	-6	-8
A	-2	1	-1	-3	-5
G	-4	-1	0	$\searrow$-2 $\downarrow$-5 $\rightarrow$-2 -2	
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G	-4	-1	0	-2	-4
C	-6	-3	-2	-1	-1 -6 -3 -1

Needleman-Wunch Implementation

- We need dynamic programming:
  - Solve smallest problems and use solutions for bigger problems
- Money change problem: minimize number of coins to make change (BA5A)
- Manhattan Tourist Problem: maximize sum of edge weights of path (longest path) (BA5B)
- Since the Manhattan Tourist Problem and Sequence Alignment are both DAGs, we can use similar dynamic programming solution (BA5C)
- But we also need the actual path to obtain the alignment!

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- Alternatives with same score:

	-	A	A	A	C
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C	-6	-3	-2	-1	-1

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Trace back/backtracking

- Trace back from last cell:
  - Need to "memorize" the decisions took every time a new cell of the S matrix is computed.
  - Store the information of which was the choice (from the three possible ones) that provided the highest score.
    - Encode direction with 3 symbol alphabet
    - build a trace-back matrix T with the same dimensions as the S matrix.
- Having the trace-back information available allows to recover the best alignment by traversing the matrix.

Needleman-Wunsch Algorithm (BA5C)

- Create matrix S of length(A)+1 by length(B)+1
- From $S_{0,0}$, fill first column and row with increasing gap penalty
- Fill matrix with max, left to right, top to bottom:
- $S_{i,j} = max \begin{cases} S_{i-1,j-1} + sm(A_i, B_j) \\ S_{i,j-1} + g \\ S_{i-1,j} + g \end{cases}$
- Trace back from last cell:
  - Diagonal: match the two residues in the alignment
  - Left: align top residue with gap
  - Up: align left residue with gap
- Ties can lead to multiple equivalent alignments

Substitution Matrices (sm)

- Protein sequences: substitution matrices are calculated based on the probabilities of finding a given pair of aligned aminoacids **in a good alignment**, compared to the probability of obtaining such a pairing by chance.
- Databases with good alignments needs to be used
- Probability of occurrence of a given pair of aminoacids is estimated by its relative frequency in the alignments.
- For each pair of aminoacids, the expected probability is estimated as if pairs were obtained by chance.

Substitution Matrices (sm)

- The score s for a given aminoacid pair a, b is given by:

$$s(a,b) = round\left(2 \times \log_2 \frac{P(a,b)}{P(a)P(b)}\right)$$

where $P(a,b)$ is the estimated probability of occurrence of the pair of aminoacids (a,b) in the database alignments, and $P(a)$ and $P(b)$ are the estimated probabilities for the occurrence of aminoacids a and b, respectively.

BLOSUM62

	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
A	**4**																			
C	0	**9**																		
D	-2	-3	**6**																	
E	-1	-4	2	**5**																
F	-2	-2	-3	-3	**6**															
G	0	-3	-1	-2	-3	**6**														
H	-2	-3	1	0	-1	-2	**8**													
I	-1	-1	-3	-3	0	-4	-3	**4**												
K	-1	-3	-1	1	-3	-2	-1	-3	**5**											
L	-1	-1	-4	-3	0	-4	-3	2	-2	**4**										
M	-1	-1	-3	-2	0	-3	-2	1	-1	2	**5**									
N	-2	-3	1	0	-3	0	-1	-3	0	-3	-2	**6**								
P	-1	-3	-1	-1	-4	-2	-2	-3	-1	-3	-2	-1	**7**							
Q	-1	-3	0	2	-3	-2	0	-3	1	-2	0	0	-1	**5**						
R	-1	-3	-2	0	-3	-2	0	-3	2	-2	-1	0	-2	1	**5**					
S	1	-1	0	0	-2	0	-1	-2	0	-2	-1	1	-1	0	-1	**4**				
T	-1	-1	1	0	-2	1	0	-2	0	-2	-1	0	1	0	-1	1	**4**			
V	0	-1	-3	-2	-1	-3	-3	3	-2	1	1	-3	-2	-2	-3	-2	-2	**4**		
W	-3	-2	-4	-3	1	-2	-2	-3	-3	-2	-1	-4	-4	-2	-3	-3	-3	-3	**11**	
Y	-2	-2	-3	-2	3	-3	2	-1	-2	-1	-1	-2	-3	-1	-2	-2	-2	-1	2	**7**

Protein Alignment (BA5E)

- Needleman-Wunsch Algorithm can also be used
- Example: PHSWG | HGWAG
  - $g = -8$
  - $sm(A_i, B_j) = blosum62(A_i, B_j)$

	gap	H	G	W	A	G
gap	0	-8	-16	-24	-32	-40
P	-8	-2	-10	-18	-25	-33
H	-16	0	-4	-12	-20	-27
S	-24	-8	0	-7	-11	-19
W	-32	-16	-8	11	3	-5
G	-40	-24	-10	3	11	9

$S_{i,j}$

	gap	H	G	W	A	G
gap	0	-8	-16	-24	-32	-40
P	-8	-2	-10	-18	-25	-33
H	-16	0	-4	-12	-20	-27
S	-24	-8	0	-7	-11	-19
W	-32	-16	-8	11	3	-5
G	-40	-24	-10	3	11	9

$T_{i,j}$

Protein Alignment

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W	-32	-16	-8	11	3	-5
G	-40	-24	-10	3	11	9

$S_{i,j}$

	gap	H	G	W	A	G
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P	-8	-2	-10	-18	-25	-33
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S	-24	-8	0	-7	-11	-19
W	-32	-16	-8	11	3	-5
G	-40	-24	-10	3	11	9

Best alignment:

```
P H S W - G
- H G W A G
```

Score of the best alignment:

-8 + 8 + 0 + 11 -8 + 6 = **9**

Summary

- Pairwise Sequence Alignment
  - We’ll see Multiple Sequence Alignments later
- Dynamic programming for aligning sequences
- Needleman-Wunch algorithm
- Scoring alignments

Further reading

- Rocha and Ferreira Chapter 6
- Compeau and Pevzner Chapter 5